

Lecture:11 Primary Memory (Main Memory)

1. Introduction to Memory in Computers

A computer's memory system is essential for storing data and instructions needed for processing. Memory is broadly classified into:

- **Primary Memory (Main Memory)** – Fast, directly accessed by CPU
- **Secondary Memory** – Slow, long-term storage
- **Cache Memory** – Very fast, small memory near CPU

Today's lecture focuses on **Primary Memory**.

2. What is Primary Memory?

Primary memory, also called **Main Memory**, is the memory that is **directly accessed by the CPU** during execution.

It is used to store:

- Program instructions currently being executed
- Data currently being processed
- Intermediate results

Primary memory acts as the **CPU's workspace**.

Key Characteristics

Feature	Description
Speed	Very fast, faster than secondary storage
Volatility	Mostly volatile (RAM) → loses data when power is off
Access	Directly accessible by CPU
Cost	More expensive per GB than secondary memory
Capacity	Smaller compared to HDD/SSD

3. Types of Primary Memory

Primary memory is mainly divided into:

1. **RAM – Random Access Memory (Volatile)**
2. **ROM – Read Only Memory (Non-volatile)**

Let's explore both in detail.

4. RAM (Random Access Memory)

RAM is the **working memory** of the computer.

When you open a program, it loads from the hard drive/SSD into RAM so the CPU can access it instantly.

4.1 Characteristics of RAM

- **Volatile:** Data is erased when power is turned off.
- **Read-Write:** Both reading and writing are possible.
- **Fast:** Much faster than secondary devices.
- **Temporary Storage:** Stores only current tasks.

4.2 Types of RAM

A. SRAM (Static RAM)

- Uses flip-flops to store data
- Faster and more expensive
- Used in **cache memory**
- No need for refresh cycles

B. DRAM (Dynamic RAM)

- Stores data in capacitors
- Requires periodic **refreshing**
- Cheaper, slower than SRAM
- Used as **main system RAM**

4.3 DRAM Subtypes

1. SDRAM (Synchronous DRAM)

- Works in sync with CPU clock
- Faster than general DRAM

2. DDR SDRAM (Double Data Rate SDRAM)

Modern computers use DDR memory.

Versions:

1. **DDR1**
2. **DDR2**
3. **DDR3**
4. **DDR4**
5. **DDR5** (latest, fastest)

Each version offers:

- Higher speed
- Lower power consumption

- Greater bandwidth

4.4 RAM Operations

When you open a program:

1. CPU retrieves program from HDD/SSD
2. Loads it into RAM
3. CPU executes instructions directly from RAM
4. Output is produced

RAM works as an **intermediary** between CPU and storage.

5. ROM (Read Only Memory)

ROM is non-volatile memory used to store **permanent instructions**.

Even when the computer is turned off, ROM retains its data.

5.1 Characteristics of ROM

- **Non-volatile**
- Used for **permanent data storage**
- Contains crucial instructions for **booting**
- Typically read-only (but modern types allow limited writing)

5.2 Types of ROM

A. PROM (Programmable ROM)

- Can be programmed once
- Cannot be erased

B. EPROM (Erasable PROM)

- Can be erased with **UV light**
- Reprogrammable

C. EEPROM (Electrically Erasable PROM)

- Can be erased electrically
- Used in BIOS/UEFI firmware

D. Flash Memory

- Advanced EEPROM
- Used in USB drives, SSDs
- Faster and more flexible

6. Difference Between RAM and ROM

Feature	RAM	ROM
Volatility	Volatile	Non-Volatile
Purpose	Temporary storage for active tasks	Permanent instruction storage
Writing	Read + write	Mostly read-only
Speed	Very fast	Slower than RAM
Examples	DDR4, DDR5	BIOS, Firmware

7. Role of Primary Memory in System Performance

A. More RAM = Better Multitasking

- Allows more programs to run simultaneously
- Reduces lag

B. RAM Speed Affects Performance

- Higher MHz → faster data throughput
- Important for gaming/video editing

C. ROM Ensures System Booting

- BIOS/UEFI stored in ROM performs POST (Power-On Self Test)
- Loads the OS into RAM

8. Memory Access Methods

Primary memory supports:

1. Random Access

- Any memory location can be accessed directly
- In the same amount of time

2. Sequential Access

- Rarely used in primary memory
- Data accessed in a sequence (used in tapes)

9. Memory Hierarchy (Advanced Concept)

Primary memory is part of the **memory hierarchy**, which optimizes speed vs cost:

1. **CPU Registers (fastest, smallest)**
2. **Cache Memory (L1, L2, L3 – very fast)**
3. **Primary Memory (RAM)**
4. **Secondary Storage (SSD/HDD)**
5. **Tertiary Storage (optical disks, tapes)**

The goal is to make the system **efficient and balanced**.

Lecture:12 Secondary Memory (Auxiliary Storage)

1. Introduction to Memory in Computers

A computer uses different types of memory to store data and instructions.

While **Primary Memory (RAM)** stores data temporarily during execution, it is:

- Volatile
- Limited in size
- Expensive

Therefore, computers need another storage system for **permanent and large-capacity data storage**. This is where **Secondary Memory** comes in.

2. What is Secondary Memory?

Secondary Memory (also called **Auxiliary or External Storage**) is used to store data:

- Permanently
- For long-term use
- In large quantities

Examples include:

- Hard Disk Drive (HDD)
- Solid-State Drive (SSD)
- USB Flash Drives
- CDs and DVDs
- Memory Cards

3. Key Characteristics of Secondary Memory

Feature	Description
Non-volatile	Data remains even when power is off
Large capacity	Stores huge amounts of data (GBs to TBs)
Lower cost per GB	Cheaper than primary memory
Slower speed	Slower than RAM and Cache
Portable options	USB drives, CDs, Portable HDDs

4. Why Secondary Memory is Needed

Primary memory alone is not enough because:

- **RAM is volatile** → data is lost after shutdown
- **RAM has limited space** → cannot store large files
- **Programs and operating systems need permanent storage**
- **Users need to save documents, images, videos, software**

Secondary memory solves all these problems.

5. Types of Secondary Memory

Secondary memory can be broadly divided into:

1. **Magnetic Storage**
2. **Optical Storage**
3. **Solid-State Storage**
4. **Cloud Storage** (modern category)

We'll discuss each in detail.

6. Magnetic Storage Devices

These devices store data using **magnetic patterns** on rotating disks or tapes.

6.1 Hard Disk Drive (HDD)

HDD is the most common magnetic storage device.

How it works

- Contains magnetic **platters** that spin at high speed
- A **read/write head** moves over the disks to store or retrieve data
- Data is stored in binary as magnetic charges

Features

- Large capacity (up to 20 TB)
- Cheaper than SSD
- Slower due to mechanical parts
- Sensitive to shock or drops

6.2 Magnetic Tapes

- Used for **backup and archival storage**
- Very large capacity

- Slow access (sequential access)

7. Optical Storage Devices

These use **laser technology** to read/write data.

Examples:

- CD (Compact Disc) — ~700 MB
- DVD (Digital Versatile Disc) — 4.7–17 GB
- Blu-Ray — 25–128 GB

How Optical Disks Work

- Laser beam reads/write data
- Data stored as **pits** and **lands** on the disk surface
- Disc spins inside drive
- Laser detects reflections representing (0s and 1s)

Types of Optical Discs

Type	Description
CD-ROM	Read-only
CD-R / DVD-R	Write once
CD-RW / DVD-RW	Rewritable
Blu-Ray (BD)	High-definition video storage

8. Solid-State Storage Devices

These devices store data in **flash memory chips**.

They have **no moving parts**, making them faster and more reliable.

8.1 Solid-State Drive (SSD)

How it works

- Uses NAND flash memory
- Data stored as electrical charges in memory cells

Advantages

- Very fast (5–10x faster than HDD)
- Durable (no moving parts)
- Low power consumption
- Silent operation

SSD Types

- **SATA SSD** — slower, older
- **NVMe SSD (M.2)** — very fast (up to 7000 MB/s)

8.2 USB Flash Drive

Also called **Pen Drive** or **Thumb Drive**.

Features

- Portable
- Uses flash memory
- Capacity: 4 GB to 1 TB
- Plug-and-play (USB port)

8.3 Memory Cards

Examples:

- SD card
- Micro SD

Used in:

- Cameras
- Phones
- Drones
- Tablets

9. Cloud Storage (Modern Secondary Storage)

Cloud storage stores data on remote servers accessed via the Internet.

Examples:

- Google Drive
- OneDrive
- iCloud
- Dropbox

Advantages

- Accessible from anywhere
- Backup and synchronization
- Massive storage capacity

Disadvantages

- Requires internet
- Security concerns
- Ongoing subscription cost

10. Comparison of Secondary Storage Types

Type	Speed	Cost	Durability	Examples
Magnetic	Medium	Low	Moderate	HDD, Tape
Optical	Low	Very low	High	CD, DVD
Solid State	High	Higher	Very high	SSD, USB
Cloud	Depends on internet	Subscription	High	Drive, iCloud

11. Access Methods

1. Sequential Access

Access data in order

Example: Magnetic tapes

2. Direct (Random) Access

- Jump directly to desired data
- SSD, USB

12. Advantages of Secondary Memory

- Permanent storage
- High capacity
- Low cost
- Portable options available
- Not volatile
- Essential for multimedia and large applications

13. Disadvantages of Secondary Memory

- Slower than RAM and Cache
- Some devices (HDD) are vulnerable to physical shock
- Optical media can scratch
- Flash memory wears out after many write cycles

14. Role of Secondary Memory in System Performance (Advanced)

A. Storage Type Matters

- SSD significantly speeds up boot time and app loading
- HDD is slower but cheaper

B. Hybrid Drives (SSHD)

- Combine SSD speed with HDD capacity.

15. Secondary Memory vs Primary Memory

Feature	Primary Memory (RAM)	Secondary Memory
Volatility	Yes	No
Speed	Very fast	Slower
Capacity	Limited	Very large
Cost	Expensive	Cheaper
CPU Access	Direct	Indirect
Use	Temporary data	Permanent storage

Lecture:13 Cache Memory (Basic to Advanced)

1. Introduction: Why Cache Memory?

The CPU is extremely fast and can execute billions of instructions per second. However, main memory (RAM) is **much slower** compared to CPU processing speed.

This mismatch creates a **bottleneck**, known as the **Memory Wall Problem**.

To solve this, computer architects introduced **Cache Memory**—a very fast, small-size memory located close to the CPU.

2. What is Cache Memory?

Cache Memory is:

- A **small, high-speed, volatile memory**
- Located very close to the CPU (often inside the CPU chip)
- Designed to temporarily store frequently accessed data and instructions
- Used to reduce the time it takes the CPU to access memory

Cache memory acts as a **bridge between the CPU and RAM**, reducing access time and improving performance.

3. Key Characteristics of Cache Memory

Feature	Description
Speed	Faster than RAM but slower than CPU registers
Volatility	Volatile (data disappears when power is off)
Size	Very small (in KB or MB)
Cost	Very expensive per MB
Location	On CPU chip or very close to CPU
Purpose	Reduce memory access time

4. Why Cache Memory is Needed (The Memory Hierarchy Problem)

CPU speed is measured in **nanoseconds**, RAM speed in **tens of nanoseconds**.

If the CPU always had to wait for RAM:

- Almost 50–70% of time would be wasted
- System performance would drop drastically

Cache reduces this delay by storing **data that will be needed soon** (temporal locality) or **data close to already accessed memory locations** (spatial locality).

5. How Cache Memory Works

Cache memory operates based on a principle called **locality of reference**:

A. Temporal Locality

If the CPU uses data once, it is likely to use it again soon.
Cache stores this data for quick access.

B. Spatial Locality

If the CPU accesses a memory location, nearby data is also likely to be used.
Cache loads blocks (called **cache lines**) instead of single bytes.

6. Cache Hit and Cache Miss

When the CPU requests data:

Cache Hit

- Data is found in cache
- Very fast
- CPU continues execution immediately

Cache Miss

- Data is NOT found in cache
- CPU must fetch it from RAM
- Slower performance

Types of cache misses:

1. **Compulsory Miss** – first-time access
2. **Capacity Miss** – cache too small
3. **Conflict Miss** – mapping issues

7. Levels of Cache Memory (L1, L2, L3)

Modern CPUs use a **multi-level cache system**.

7.1 L1 Cache (Level 1)

- Smallest (usually 32–64 KB per core)
- Fastest (located inside the CPU core)
- Divided into:
 - **L1 Instruction Cache (I-Cache)**
 - **L1 Data Cache (D-Cache)**

Purpose:

Store most frequently accessed instructions and data.

7.2 L2 Cache (Level 2)

- Larger than L1 (128 KB–1 MB)
- Slightly slower
- Can be private to each core or shared

Purpose:

Store additional data not found in L1.

7.3 L3 Cache (Level 3)

- Much larger (4–64 MB)
- Slower than L1 and L2
- Shared among all CPU cores

Purpose:

Reduce RAM access for all cores and improve multicore performance.

8. Types of Cache Based on Mapping

A. Direct-Mapped Cache

- Each memory block maps to **exactly one** cache line
- Simple but prone to conflict misses

B. Fully Associative Cache

- Memory block can go **anywhere** in cache
- Very flexible but expensive

C. Set-Associative Cache

- Cache divided into sets
- Each block can go to **any line within a set**
- Best balance of speed and cost

Most modern CPUs use **set-associative caches**.

9. Write Strategies in Cache

When CPU writes data:

A. Write-Through

- Data written to cache AND RAM simultaneously
- Safer but slower

B. Write-Back

- Data written only to cache at first
- RAM updated later
- Faster but needs tracking (dirty bit)

10. Cache Line and Block Size

Cache stores data in small fixed-sized chunks called **cache lines** (32, 64, or 128 bytes). Large cache lines improve spatial locality but may increase miss penalty.

11. Replacement Policies

When cache is full and new data must be added:

- **LRU (Least Recently Used)** → replaces the oldest used data
- **FIFO (First In First Out)**
- **Random Replacement**

LRU is most common.

12. Inclusive vs Exclusive Cache

Inclusive Cache

$L1 \subseteq L2 \subseteq L3$

- Easier synchronization
- Uses more storage

Exclusive Cache

Caches hold **different** data

- More total data stored
- Complex design

13. Importance of Cache in Modern CPUs

Cache significantly improves performance by reducing average memory access time (AMAT).

Formula:

$$\text{AMAT} = \text{Hit Time} + \text{Miss Rate} \times \text{Miss Penalty}$$

Lower miss rate → higher performance.

14. Cache Memory Size in Real CPUs (Examples)

Intel Core i7

- L1: 64 KB per core
- L2: 256 KB per core
- L3: 8–16 MB shared

15. Advantages of Cache Memory

- Very fast data access
- Reduces RAM access frequency
- Improves CPU execution speed
- Reduces power consumption
- Essential for multitasking and gaming performance

16. Disadvantages of Cache Memory

- Very costly
- Limited size
- Complex architecture
- Can introduce consistency issues in multicore systems